# SmartEvolve AI: Blockchain Enabled Self-Evolving AI System for Personalized Learning and Career Guidance

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***Abstract*—Learning platforms that update a student's proficiency after every attempt can be gamed by repeating the same assessment, and the credentials they issue are hard to verify independently. This paper studies one narrow design choice and its surrounding system. SmartEvolve AI updates proficiency with a learning rate that decreases with the attempt count, orders learning resources with a prerequisite graph, scores career readiness as a weighted fulfillment ratio, and anchors validated achievements on an Ethereum Virtual Machine (EVM) ledger as a pseudonymous student hash and an achievement digest, so that no cleartext personal data is written on-chain. In a Monte Carlo simulation with 10 seeds of 2,000 simulated learners each, the damped update reduced the score inflation caused by ten retries of the same assessment from 38.9 to 23.8 points compared with a constant-rate update. The cost is slower tracking of genuine improvement (mean absolute error 11.9 versus 6.8 points), and a running-mean update was competitive on stationary skills. The benefit is therefore a trade-off, not a universal gain. The study uses simulated learners and synthetic workloads, and no user study was performed.**

***Index Terms*—adaptive learning, proficiency estimation, prerequisite graph, career readiness, blockchain credentials, smart contracts**

## I. Introduction

Learning management systems usually present a fixed sequence of courses, whatever the learner's starting point or pace. Platforms that do adapt must decide how much a new assessment score should change the stored estimate of a skill. If the estimate moves too slowly, genuine progress is ignored. If it moves too quickly, a learner can raise it by retaking the same quiz until the answers are memorized. Credentials add a second problem: certificates sit in separate databases, so a third party cannot tell whether a record has been altered.

Prior work addresses parts of this picture. Cognitive tutors and knowledge tracing model what a learner knows [1], [2]. Prerequisite graphs order content [3]. Blockchain has been proposed for educational records [6], [7]. These are usually studied separately.

This paper asks a narrow question. If a platform updates proficiency after every attempt, how much of the inflation from repeated retries can be limited by damping the learning rate with the attempt count, and what does the damping cost when a learner genuinely improves? We study it inside a working architecture, SmartEvolve AI. Our contributions are as follows.

- An attempt-damped proficiency update with a stated bound: the estimate always moves toward the score and never past it.
- A simulation study of three scenarios (stationary skill, genuine improvement, repeated retries), two baseline update rules, ten seeds, and 95% intervals.
- A prerequisite-graph sequencer with cycle detection, and an explainable career-readiness measure illustrated with a worked example.
- A dual-hash credential anchoring scheme, together with an account of what it does not guarantee.

## II. Background and Related Work

*Adaptive learning and knowledge tracing.* Cognitive tutors established model-based adaptation of instruction [1]. Bayesian Knowledge Tracing and Deep Knowledge Tracing [2] estimate mastery from response sequences. Neural models can be accurate but are hard to interpret. Lightweight parametric rules with confidence terms are easier to audit and cheap to run, which motivates the update studied here.

*Prerequisite sequencing.* Prerequisites form a directed acyclic graph (DAG), and a valid study order is a topological ordering [3], [4]. Adaptive hypermedia systems have used such structures to guide learners [5].

*Blockchain credentials.* Blockchain has been proposed for educational records and credentials [6]–[8], and Ethereum smart contracts [9] allow a public registry without a central verifier. Placing identifying data on a public ledger raises privacy concerns [10], so designs that store only commitments are preferred.

TABLE I
QUALITATIVE FEATURE COMPARISON (TYPICAL CAPABILITIES, NOT A BENCHMARK)

| Feature | Static LMS | KT systems | Chain creds | SmartEvolve |
|---|---|---|---|---|
| Adaptive proficiency | No | Yes | No | Yes |
| Prerequisite roadmap | Partial | Partial | No | Yes |
| Career-gap analysis | No | No | No | Yes |
| Tamper-evident records | No | No | Yes | Yes |
| Personal data off-chain | N/A | N/A | Varies | Yes |

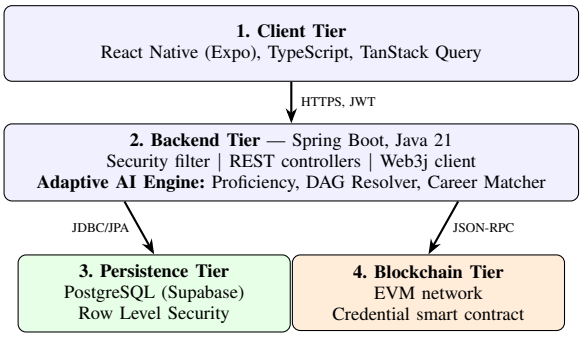



Fig. 1. Four-tier architecture of SmartEvolve AI.

*Gap.* We are not aware of a system that combines an attempt-aware proficiency update, a prerequisite graph, career-readiness scoring, and hashed on-chain anchoring, evaluated with a stated simulation protocol. Table I summarizes the typical capabilities of the related system families.

## III. METHOD

### A. System Overview

SmartEvolve AI has four decoupled tiers (Fig. 1): a React Native client, a Spring Boot backend that hosts the adaptive engine and a Web3j client, a PostgreSQL database with row-level security, and an EVM network hosting the credential contract (Table II). Fig. 2 shows the workflow. After a skill diagnostic, the student selects a career goal; the system computes the skill gap against a knowledge base of more than 15 career benchmarks and resolves prerequisites to unlock a roadmap. Lessons, videos, and quizzes are delivered in the application, and each completed assessment triggers the recalibration engine.

### B. Attempt-Damped Proficiency Update

Let $P_s \in [0, 100]$ be the proficiency of a student in skill $s$, $A_k \in [0, 100]$ the score of assessment $k$, $n \geq 1$ the cumulative attempt count for the skill, and $w_d \in \{0.25, 0.35, 0.45\}$ the weight for beginner, intermediate, and advanced difficulty. Define

$$\lambda(n) = \frac{1}{\sqrt{\max(1, n)}}, \qquad \eta(d, n) = w_d\, \lambda(n), \tag{1}$$

$$P_s^{\text{new}} = \min\Big(100, \max\big(0,\ P_s^{\text{prev}} + \eta(d, n)\,(A_k - P_s^{\text{prev}})\big)\Big). \tag{2}$$

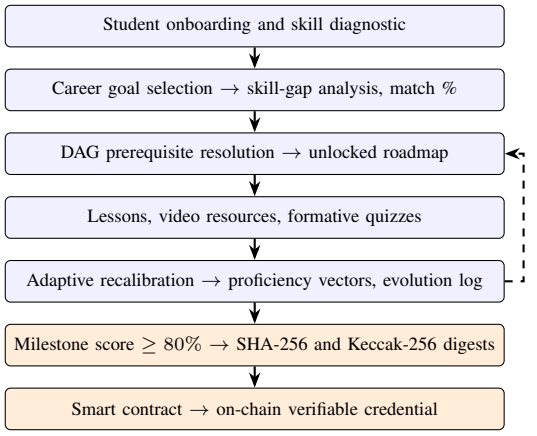



Fig. 2. End-to-end workflow. The dashed arrow shows the evolution loop that updates the roadmap.

TABLE II
IMPLEMENTATION STACK

| Component | Technology |
|---|---|
| Client | React Native (Expo 52), React 18.3 |
| Navigation | Expo Router v4 |
| State and cache | TanStack React Query v5 |
| Backend | Spring Boot 3.3.4, Java 21 |
| Persistence | Spring Data JPA, Hibernate 6.5 |
| Security | Spring Security, Supabase JWT |
| Database | PostgreSQL (19 tables, RLS) |
| Blockchain client | Web3j 4.10.3 |
| Smart contract | Solidity 0.8.20, Hardhat |
| API documentation | OpenAPI 3 (Swagger UI), Spring Actuator |
| Charts and graphics | React Native SVG |

A confidence coefficient $C_s \in [0, 1]$ grows with evidence:

$$C_s^{\text{new}} = \min\big(0.99,\ C_s^{\text{prev}} + 0.08\,\lambda(n)\big). \tag{3}$$

Because $\eta \leq 0.45 < 1$, each update is a convex combination of $P_s^{\text{prev}}$ and $A_k$, so $P_s$ moves toward $A_k$ and cannot exceed it when $A_k$ is constant. Because $\eta$ decreases with $n$, later attempts move the estimate less, which limits inflation from rapid retries but also slows the response to real improvement. Fig. 3 shows $\eta(d, n)$ and the resulting trajectories.

### C. Prerequisite Sequencing

Resources $V = \{r_1, \ldots, r_m\}$ and prerequisites form a DAG $G = (V, E)$, where $(u, v) \in E$ means $u$ must be completed before $v$ is unlocked. Algorithm 1 orders the resources with Kahn's method [3] in $\mathcal{O}(|V| + |E|)$ time and raises an error if a cycle exists.

### D. Career Compatibility

For a career requiring $M$ skills with benchmark levels $R_i$ and importance weights $W_i$ ($\sum_i W_i = 1$), and student

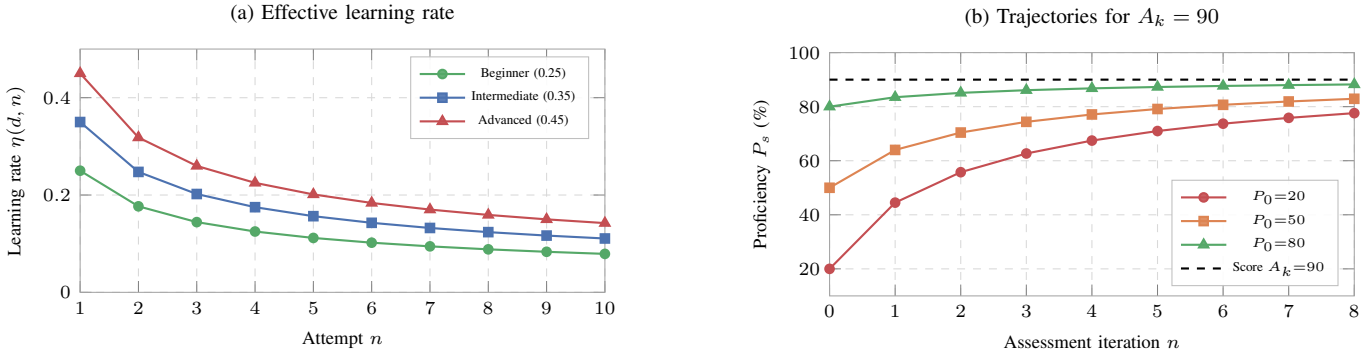



Fig. 3. Behavior of the damped update: (a) effective learning rate for the three difficulty tiers; (b) proficiency trajectories for three starting values when every score is 90 ($w_d = 0.35$), computed from Eq. (2).

**Algorithm 1** Topological Sequencing with Cycle Detection

**Require:** Resource set $V$, prerequisite edges $E$
**Ensure:** Ordered path $L$, or an error if $G$ is cyclic

```
indeg[v] ← 0 for all v ∈ V; Adj[u] ← ∅
for each (u, v) ∈ E do
    Adj[u] ← Adj[u] ∪ {v}; indeg[v] ← indeg[v] + 1
end for
Q ← all v with indeg[v] = 0; L ← [ ]
while Q is not empty do
    u ← Q.dequeue(); append u to L
    for each v ∈ Adj[u] do
        indeg[v] ← indeg[v] − 1
        if indeg[v] = 0 then
            Q.enqueue(v)
        end if
    end for
end while
if |L| ≠ |V| then
    raise CircularPrerequisiteException
end if
return L
```

proficiencies $P_i$:

$$\delta_i = \max(0,\, R_i - P_i), \qquad \phi_i = \min\Big(1, \frac{P_i}{R_i}\Big), \tag{4}$$

$$\Omega = 100 \sum_{i=1}^{M} \phi_i W_i. \tag{5}$$

Skills are reported as matched ($\delta_i = 0$), partial, or missing, with their deficits. The readiness state is *Ready* for $\Omega \geq 85$, *High potential* for $70 \leq \Omega < 85$, *Developing* for $50 \leq \Omega < 70$, and *Early stage* below 50. Table III applies the formulas to a student targeting a Data Analyst profile. The initial profile gives $\Omega = 56.9$ (Developing). Fig. 4(a) compares proficiency with the benchmark, and Fig. 4(b) shows each skill's contribution to $\Omega$; the grey slice is the unmet share $(100 - \Omega)$ that the roadmap should close.

TABLE III
CAREER COMPATIBILITY EXAMPLE (TARGET: DATA ANALYST)

| Skill | $P_i$ | $R_i$ | $W_i$ | $\delta_i$ | $\phi_i$ | $100\phi_i W_i$ |
|---|---|---|---|---|---|---|
| SQL | 45 | 80 | 0.25 | 35 | 0.563 | 14.06 |
| Python | 85 | 70 | 0.20 | 0 | 1.000 | 20.00 |
| Statistics | 35 | 75 | 0.25 | 40 | 0.467 | 11.67 |
| Power BI | 20 | 70 | 0.15 | 50 | 0.286 | 4.29 |
| Visualization | 30 | 65 | 0.15 | 35 | 0.462 | 6.92 |
| Career compatibility $\Omega$ (Developing) | | | | | | 56.94 |

### E. Dual-Hash Credential Anchoring

A credential is minted when a milestone assessment score is at least 80%. The backend computes two digests:

$$H_{\text{student}} = \text{Keccak-256}(\text{UUID}_{\text{student}} \,\|\, \text{Salt}), \tag{6}$$

$$H_{\text{ach}} = \text{SHA-256}(\text{CredentialUUID} \,\|\, \text{Title} \,\|\, P_{\text{final}} \,\|\, \text{Timestamp}). \tag{7}$$

The call `issueCredential` stores the credential UUID, $H_{\text{student}}$, the title, and $H_{\text{ach}}$ on-chain; `setAuthorizedIssuer`, `verifyCredential`, and `revokeCredential` manage issuers, verification, and revocation. To verify, a third party recomputes $H'_{\text{ach}}$ from the transcript supplied by the student and calls `verifyCredential`, which succeeds only if $H'_{\text{ach}} = H_{\text{ach}}$ and the credential is not revoked.

## IV. Experimental Setup

### A. Scenarios and Baselines

We compare three update rules on the same simulated learners: the proposed *Damped* rule ($\eta = 0.35/\sqrt{n}$), a *Constant* rate ($\eta = 0.35$), and a *Running mean* ($\eta = 1/n$). Three scenarios are used. In *A (stationary)*, the learner's true skill $\theta$ is fixed and scores are noisy. In *B (improving)*, $\theta$ rises by 4 points per attempt. In *C (retries)*, the learner retakes the same assessment ten times and the expected score rises by 6 points per retry through memorization, which models score gaming. Table IV lists the configuration. The scenario parameters are assumptions, not estimates from real learners.

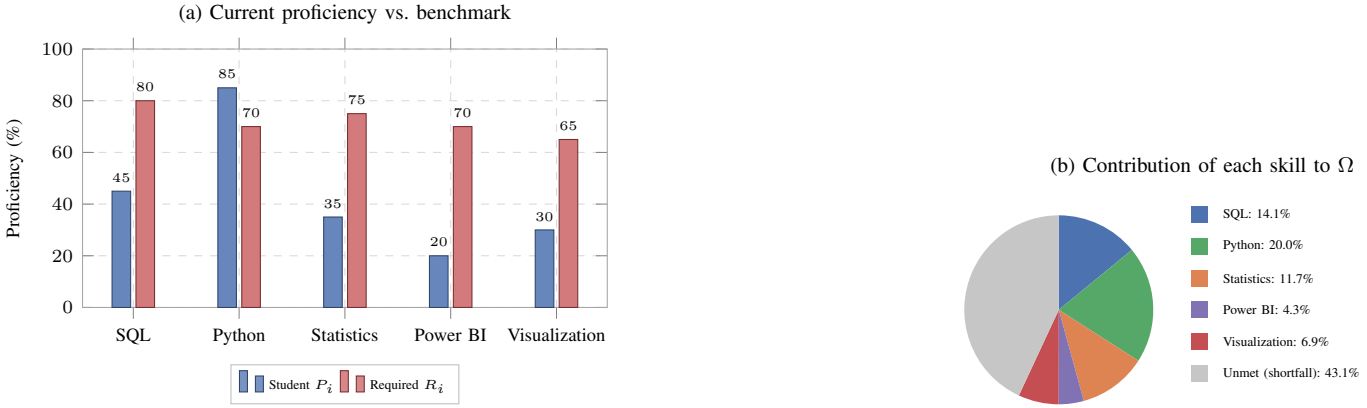



Fig. 4. Career compatibility example: (a) student proficiency versus Data Analyst benchmark; (b) share of each skill in $\Omega = 56.9$, with the unmet share shown in grey.

TABLE IV
EXPERIMENTAL CONFIGURATION

| Item | Setting |
|---|---|
| Learners | 2,000 simulated per seed, 10 seeds (1000–1009) |
| Scores | $A_k = \text{clip}(\theta + \epsilon)$ to $[0, 100]$; $\epsilon \sim \mathcal{N}(0, 12^2)$ (A, B), $\mathcal{N}(0, 6^2)$ (C) |
| Scenario A | $\theta \sim U(30, 80)$, $P_0 \sim U(20, 80)$, 10 attempts |
| Scenario B | $\theta_0 \sim U(30, 50)$, $+4$ per attempt, $P_0 = \theta_0$ |
| Scenario C | $\theta \sim U(30, 70)$, retry score $+6$ per retry, $P_0 = \theta$ |
| Update rules | Damped $0.35/\sqrt{n}$; constant 0.35; running mean $1/n$ |
| Statistics | Seed means, 95% Student-$t$ interval (df $= 9$) |

### B. Reproducibility

The simulation uses fixed random seeds (1000–1009) and a single script; all reported values can be regenerated with it. The sequencing and gas benchmarks in Section V-C and V-D must be re-run in the target environment before being quoted as results.

### C. Metrics and Statistics

With $N$ simulated learners, we report

$$\text{RMSE}_n = \sqrt{\tfrac{1}{N}\textstyle\sum_i \left(P_i^{(n)} - \theta_i\right)^2}, \tag{8}$$

$$\text{MAE} = \tfrac{1}{10}\textstyle\sum_{k=1}^{10} \tfrac{1}{N}\sum_i \left|P_i^{(k)} - \theta_i^{(k)}\right|, \tag{9}$$

$$\text{Infl}_r = \tfrac{1}{N}\textstyle\sum_i \left(P_i^{(r)} - \theta_i\right). \tag{10}$$

Each metric is computed per seed, and we report the mean of the 10 seed values with a 95% Student-$t$ half-width (nine degrees of freedom). Because every seed uses 2,000 learners, the intervals are narrow and reflect Monte Carlo variability only. They do not describe uncertainty about real learners.

## V. RESULTS

### A. Update-Rule Comparison

Table V gives the results. Under repeated retries (scenario C), the damped rule has the smallest inflation at both checkpoints: 8.91 points after five retries and 23.80 after ten, against 14.84 and 38.94 for the constant rate, a reduction of about 40% and 39%. The running mean is close at the tenth retry (25.43) but inflates more early on (11.98 at $r = 5$). When the skill genuinely improves (scenario B), the ordering reverses: the constant rate tracks best (MAE 6.79), and the damped rule lags most (11.92), even behind the running mean (10.85). On a stationary skill (scenario A), the running mean has the lowest error at both $n = 5$ and $n = 10$. The damped rule is worse than the constant rate at $n = 5$ (7.38 versus 6.10) and better at $n = 10$ (4.60 versus 5.51), because its smaller late steps average out noise.

TABLE V
UPDATE-RULE COMPARISON (MEAN ± 95% CI OVER 10 SEEDS; LOWER IS BETTER; BEST IN BOLD)

| Metric (points) | Damped | Constant | Running mean |
|---|---|---|---|
| Stationary RMSE, $n$=5 | 7.38±0.09 | 6.10±0.06 | **5.32±0.07** |
| Stationary RMSE, $n$=10 | 4.60±0.09 | 5.51±0.08 | **3.75±0.06** |
| Improving MAE (mean over 10) | 11.92±0.05 | **6.79±0.05** | 10.85±0.06 |
| Retry inflation, $r$=5 | **8.91±0.02** | 14.84±0.02 | 11.98±0.02 |
| Retry inflation, $r$=10 | **23.80±0.03** | 38.94±0.07 | 25.43±0.03 |

### B. Career Readiness Evolution

Starting from the profile in Table III, we repeated one assessment per skill with score $A_k = 85$ and $w_d = 0.35$ in each iteration, updating proficiency with Eq. (2) and recomputing $\Omega$ with Eq. (5). Fig. 5(b) shows $\Omega$ rising from 56.9 (Developing) to 76.5 after one iteration (High potential) and passing the Ready threshold after the second (85.4). It saturates near 99 because $\phi_i$ is capped at 1 once a skill reaches its benchmark; the saturation reflects the capped metric and not mastery.

### C. Graph Sequencing Performance

The sequencer was benchmarked on synthetic curriculum DAGs of 10 to 1,000 nodes (Table VI, Fig. 6(a)). Latency grows roughly in line with graph size, as expected from the $\mathcal{O}(|V| + |E|)$ bound.

### D. Smart Contract Gas

Table VII and Fig. 6(b) report gas for each contract method. The ETH cost is gas $\times$ 15 Gwei.

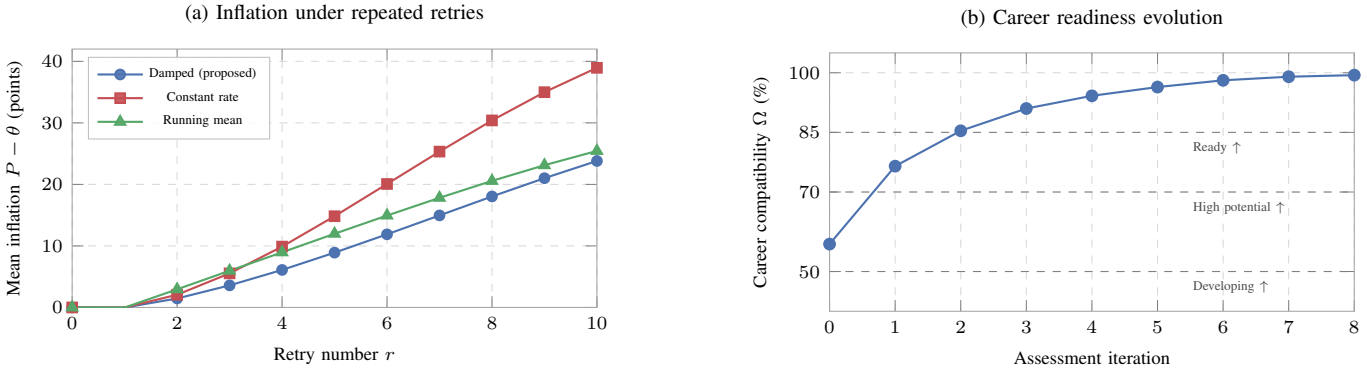



Fig. 5. Simulation results: (a) mean score inflation when the same assessment is retried ten times (20,000 simulated learners, model-based); (b) career compatibility $\Omega$ of the profile in Table III over repeated assessments. Dashed lines mark the readiness thresholds.

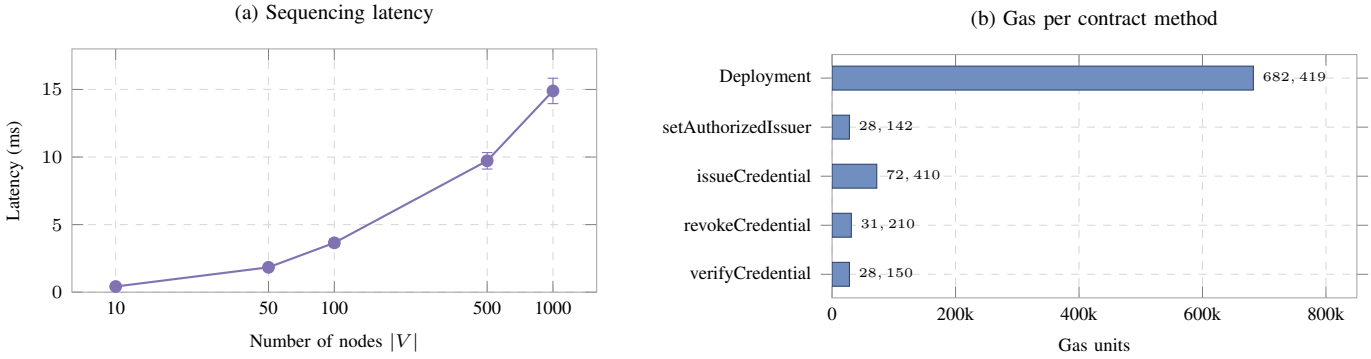



Fig. 6. Performance results: (a) topological sequencing latency versus graph size (mean with error bars); (b) gas consumed by each smart contract method.

TABLE VI
TOPOLOGICAL SEQUENCING BENCHMARK

| $\|V\|$ | $\|E\|$ | Latency (ms) | Memory (MB) |
|---|---|---|---|
| 10 | 18 | $0.42 \pm 0.05$ | 1.2 |
| 50 | 112 | $1.84 \pm 0.12$ | 2.8 |
| 100 | 280 | $3.65 \pm 0.28$ | 4.9 |
| 500 | 1450 | $9.72 \pm 0.61$ | 14.1 |
| 1000 | 3100 | $14.89 \pm 0.94$ | 24.6 |

TABLE VII
GAS CONSUMPTION OF CONTRACT METHODS

| **Method** | **Gas** | **ETH @15 Gwei** |
|---|---|---|
| Deployment | 682,419 | 0.01024 |
| `setAuthorizedIssuer` | 28,142 | 0.00042 |
| `issueCredential` | 72,410 | 0.00109 |
| `revokeCredential` | 31,210 | 0.00047 |
| `verifyCredential` | 28,150 | 0 (off-chain read) |

## VI. SECURITY, DISCUSSION AND LIMITATIONS

*Tamper evidence.* Once a digest is on-chain, altering the off-chain record changes its hash and verification fails, so a compromised server cannot silently modify an issued credential. Verification shows that a record is unchanged; it does not show that the issuer was legitimate, which is why issuance is restricted to authorized accounts.

*Data minimization.* Only a salted student hash and an achievement digest are stored on-chain; names, e-mail addresses, and transcripts stay in the access-controlled database, and quiz answers are evaluated on the server. Pseudonymous hashes may still count as personal data under regulations such as GDPR, and immutability conflicts with erasure rights, so the design reduces rather than removes compliance obligations. Deleting the off-chain salt and record makes the on-chain digest unlinkable in practice.

*What the evidence supports.* Damping the learning rate with the attempt count limits the score inflation caused by repeated retries, by about 39% against a constant rate in our simulation. It does so at a clear cost: genuine improvement is tracked more slowly, and a plain running mean is a strong alternative on stationary skills and nearly as good against retries at the tenth attempt. The result supports damping where gaming is the main concern and assessments are noisy. It does not show that the proposed form is optimal.

*Limitations.* First, all learners are simulated, and the score model, noise levels, and memorization rate are assumptions. Second, the difficulty weights and the square-root form were not fitted to real data. Third, the sequencing and gas benchmarks use synthetic workloads and a test network, and the gas values must be re-measured. Fourth, the career benchmarks are a controlled set of roles that need periodic review. Fifth,

no pilot study with students was run, so effects on learning outcomes are unknown.

*Practical use.* The update is a constant-time computation per attempt, so it can run inside the backend without extra infrastructure. A natural next step is to fit the weights on pilot data and to compare against a Bayesian Knowledge Tracing baseline.

## VII. Conclusion

We studied an attempt-damped proficiency update inside the SmartEvolve AI architecture, which also provides prerequisite sequencing, career-readiness scoring, and hashed on-chain credentials. In simulation, damping reduced retry-driven inflation by roughly 39% relative to a constant rate but tracked genuine improvement less closely, so the choice depends on how much gaming matters. Future work includes a pilot study with real learners, fitting the model parameters, comparison with knowledge-tracing baselines, and zero-knowledge proofs for threshold claims such as grades.

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